NAJEEB ABDI

Computer Science Student | Applied AI Engineer

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EDUCATION

B.S. in Computer Science — *Expected May 2027*
Qatar University | Focus: Applied AI & Machine Learning — *Doha, Qatar*

EXPERIENCE

Industry Partnership Intern, Scale AI — *Jun 2026 – Aug 2026*
Qatar University Industry Partnership Program — *Doha, Qatar*

- Engineered the CS Intake Planner end to end: an agent-based Monte Carlo simulation of QU CS students moving through the real 41-course study plan (prerequisites, seat limits, pass/fail, dropout), built in Python and FastAPI.
- Turned the simulation into a decision tool that recommends the largest intake within department-set limits, with 95% uncertainty bounds, delivered as a Next.js dashboard for the Head of Department, with a local-LLM advisor (Ollama) that explains results.

AI and Automation Intern — *Oct 2025 – Jan 2026*
Silatha (Remote) — *Amsterdam, Netherlands*

- Designed and deployed 3 production AI automation workflows (n8n, Python, OpenAI API) with API integrations, monitoring, and error handling, replacing manual processes.
- Built and deployed a 6-level educational word search game (React, Firebase) with cross-device progress sync, played by ~200 players.

Research Intern, AI-Based Aquaponics System Development — *Aug 2025 – Nov 2025*
Qatar University — *Doha, Qatar*

- Joined a multi-institution team building a federated aquaponics control system; set up its reinforcement learning model.

PROJECTS

Thermal Trace, Co-Founder & Lead Developer | Open Innovation Award, UN Reboot the Earth Hackathon — *Sep 2026 – Present*
Cold-chain decision platform for Qatar/GCC (FastAPI, MQTT, PostgreSQL, Redis, React) — *Doha, Qatar*

- Led development in a 5-person team; built the Arrhenius shelf-life model that turns temperature history into remaining safe life.
- Enforced a "deterministic math owns every number" architecture: every value carries provenance, and the LLM only explains.
- Designed the end-to-end system: truck telemetry over MQTT into FastAPI, with ML spoilage-risk estimates and a cold-store optimizer.

CodeTutor, Solo Developer — *Sep 2026 – Present*
AI coding tutor, auto-grader, and eval lab for intro CS (FastAPI, Postgres/pgvector, Next.js, Docker) — *Doha, Qatar*

- Built a sandboxed auto-grader and a 33-problem Python bank with hidden tests, validated against 97 known-wrong solutions; the same grader scores student code, evaluates LLMs (Qwen2.5-Coder 1.5B and 7B, pass@k with intervals), and is the reward for GRPO training.
- Built a local-LLM hint tutor with hybrid retrieval (right note section in the top 3 for 40 of 40 test questions) and a guardrail against leaking solutions; reported results honestly, including a GRPO gain within noise.

Lectura, Solo Developer — *Mar 2026 – Oct 2026*
AI study tool that turns lecture videos into timestamped notes (FastAPI, faster-whisper, Next.js, Supabase, Docker) — *Doha, Qatar*

- Built an AI pipeline that turns lecture videos into timestamped notes: yt-dlp download, local faster-whisper transcription, chunking, and map-reduce LLM summarization with provider fallback across Groq, OpenAI, and local Ollama, plus optional slide OCR.
- Wrapped it in an authenticated FastAPI service (Supabase auth, row-level security, free-tier quotas of 3 videos/month, live progress streaming) with a Next.js frontend (timestamps, export, "chat with your notes"), containerized with Docker; tested on 5 lecture videos from 3 to 30 minutes, all processed successfully.

RouteyAI, Founder & Lead Developer — *Mar 2026 – Sep 2026*
School bus route planning and live tracking for Qatar schools — *Doha, Qatar*

- Built solo: K-means clustering plus nearest-neighbour and cheapest-insertion routing on Mapbox driving times, and web and mobile apps for admins, drivers, and parents (Next.js, Expo, Supabase/PostGIS).
- Wound down in Sep 2026 after the Ministry of Education launched its own national attendance system.

ACTIVITIES & LEADERSHIP

Member, Events Committee, Artificial Intelligence Organization — *Apr 2025 – Present*
Qatar University — *Doha, Qatar*

- Help organize campus-wide AI events, including AI Week and the Innovation Program, coordinating logistics and demo setup.

Participant and Ambassador, Shell Intilaaqah Entrepreneurship Program — *Sep 2025 – Present*
Shell Qatar — *Doha, Qatar*

- Pitched an early-stage startup concept under mentor guidance; hosted (MC) the Fall 2026 masterclass "AI-Powered Innovation".

Also: Member, Events Team and Tech Team, ACM QU; Events Member, Mechatronics Committee; Student Assistant, IT Student Task Force.

SKILLS

AI / ML: Machine Learning, Deep Learning (TensorFlow), Reinforcement Learning, Federated Learning, Monte Carlo Simulation, LLM Apps (OpenAI, Groq, Ollama), RAG, LLM Evals, GRPO, Whisper / OCR, n8n
Web & Full-Stack: React, Next.js, TypeScript, Vite, Astro, Expo, FastAPI, Flask, REST APIs, WebSockets, Tailwind CSS
Data & Infrastructure: Supabase (PostgreSQL, PostGIS, Realtime, RLS), Firebase, Redis, MQTT, Mapbox GL JS
Languages & Tools: Python, JS/TS, SQL, Java, HTML/CSS, Git, pytest, Playwright, GitHub Actions, Docker | **Spoken:** English (Fluent), Arabic